

From Human-Machine Co-Creativity to Distributed Creativity: The Case of AI Integration in Filmmaking

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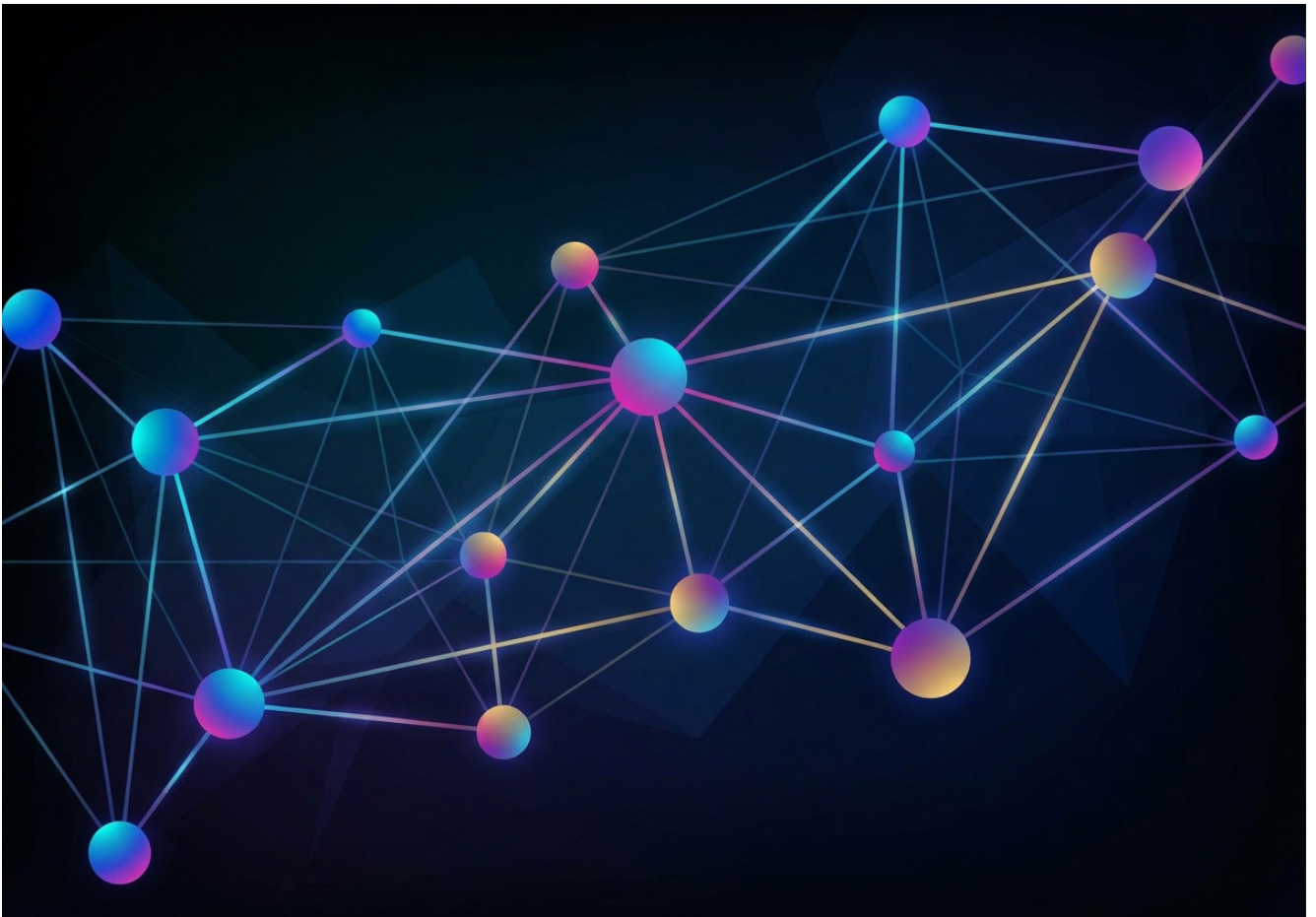
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ABSTRACT

Recent developments in Generative AI technologies have raised questions about their role in creative processes. Some scholars hold that they do exhibit creativity, others stress that there is a strong interaction among humans and machines, and talk about co-creativity. The limit of both approaches is that they first address the question of what creativity is and then apply it to the process under study. Consequently, they presuppose how the creative process works before even observing it. To investigate the matter empirically, in line with Science and Technology Studies, one has to move from what to where creativity is. One has to look at the artifacts, practices, and socio-economic arrangements that give rise to artistic products. To advance our argument, we will focus on audiovisual production – specifically Filmmaking – as a key case study. We will provide an overview of some AI techniques and show how they might be used to overcome common problems met in film shootings and editing with an impact on film aesthetics. Following Vlad Glăveanu (2014), audiovisual work is not an individual act: film aesthetics is not just due to the director and their vision but the effect of the interactions among all the people (cinematographers, editors, actors) and the technologies (lenses, tripods, lighting) involved in the process. Indeed, the very technologies shape and constrain aesthetic choices filmmakers make. New technologies (sound, color, digital) have always redefined working practices in Filmmaking eventually leading to a collapse of the work-flows. Hence, AI tools represent a technological shift that can reshape existing creative practices – both affording and constraining aesthetic and working possibilities. Ultimately, we argue that AI's impacts on creative processes demand examination beyond technical functioning, requiring instead a situated analysis of how these technologies became part of everyday creative practices.



Introduction

Recently, the discourse surrounding AI and creativity has shifted from debating the autonomous capabilities of machines toward an analysis of human-AI interactions. This transition toward ‘computational co-creativity’ emphasizes the collaboration between human and artificial agents in constructing shared artifacts (Rafner, 2025). We hold this approach to be limiting as it often overlooks the situated interactions among the diverse actors, both human and non-human, involved in the creative process, relegating such elements to the background of the AI-human interaction. Drawing on Science and Technology Studies (STS), we argue that these aspects should instead be considered as part of the actual creative practices involving AI systems. These observations are particularly evident in the debate over GenAI in Filmmaking, where claims of labor replacement or augmentation frequently fail to investigate the actual dynamics of production practices.

This paper seeks to overcome these limitations by proposing a distributed creativity framework. Through an STS methodology, we challenge current assumptions to demonstrate how creativity is distributed across actors, spaces, and temporalities, providing a perspective that better explains how AI innovations reconfigure working practices and film aesthetics.

Aims and objectives

The purpose of the present work is two-fold. First, it seeks to move beyond the limitations of the current discourse on AI and co-creativity by proposing an alternative approach rooted in the notion of distributed creativity. In doing so, we challenge four prevalent assumptions by grounding the debate in the historical development of Filmmaking practices. Secondly, building on this theoretical foundation, we argue that the impact of GenAI in Filmmaking is best understood by examining how specific applications

Cover Image caption

are integrated into existing practices. Our analysis focuses on the manipulation and generation of media assets (images, sounds, and video) to trace their effects on production pipelines. Through a taxonomy of AI's effects on assets, we discuss how AI could produce specific and important sociomaterial reconfigurations of Filmmaking.

Methods and/or procedure

Our methodology is structured around three interconnected phases:

- **Theoretical Problematization and Reframing.** We critique the most prominent frameworks in the AI co-creativity discourse and deconstruct four prevalent assumptions that limit current understanding of AI integration into creative practices. Building on this critique, we establish a new set of methodological orientations for studying 'creativity in the making', grounded in the notion of distributed creativity.
- **Application Domain and Historical Context.** The conceptual lens is applied to Filmmaking to demonstrate how it is a distributed process where agency is shared across professionals, spaces, and temporalities. By integrating film and media studies literature, we show how past technological innovations, long before GenAI, have historically reconfigured production workflows, establishing the contextual precedent for the current integration of GenAI.
- **Taxonomy Construction.** Drawing from the theoretical shifts identified in the previous phases, we develop a tentative taxonomy of AI techniques for Filmmaking. This will be used as an analytical tool to show the sociomaterial reconfiguration of practices, professional roles, and film aesthetics.

Theoretical Problematization and Reframing

Co-creativity research synthesizes long-standing debates in computational creativity with Human-Computer Interaction (Davis & Rafner, 2025). A co-creativity system is defined as "a system involving at least one human agent and one

artificially intelligent agent, collaborating with each other to build shared creative artifacts" (Rafner, 2025, p. 821). The main difference between considering 'autonomous' creative systems, which do not interact with humans during the creative processes, and co-creative systems is the central role of the interactions between the human and the AI system (Rezwana & Maher, 2022). Several frameworks have been proposed for designing and evaluating these interactive aspects (Kantosalo & Takala, 2020; Rezwana & Maher, 2022; Lin et al., 2023; Davis et al., 2025). We argue that the current debate is significantly limited by common assumptions underlying these frameworks, and we address four of them: a) the persistent description of creativity in cognitive terms; b) the narrow analytical focus on the human-AI dyad; c) the exceptional agency attributed to AI; d) the ontological approach that attempts at defining 'what is creativity'.

First, many frameworks describe the co-creative process primarily in cognitive terms, drawing heavily from cognitive science and psychological theories. For example, the Co-Creative Design Framework "incorporates concepts from enactive and embodied cognition" (Davis et al., 2025, p. 564). Also, Rezwana & Maher define the creative product in cognitive terms as "the idea or concept that is being created" (2022, p. 11). This cognitive framing on creativity neglects the sociomaterial aspects that define all creative processes since it does not consider neither the feedback between the creators and their artifacts nor the network of interobjective relations (Latour, 1996) that contribute to the creative product—i.e. the interactions between AI and other technical artifacts. Secondly, interaction is often conceived as a closed feedback between the human and the AI, focusing on their direct relationship and the shared artifact, creating a sharp dichotomy between the 'core' interaction and the 'context'. While some frameworks do acknowledge the material context, they often treat it as an external factor. The 'Five C's' framework, for instance, includes a context dimension, the 'material surroundings of the work', which is treated as external layers that influence the core collaboration rather than being part of the creative process itself (Kantosalo & Takala, 2020). From an STS perspective, the boundary between an action

and its background should not be treated as a pre-given, but as an outcome of practice that requires explanation (Callon, 1998; Latour, 2005). This act of framing is not merely conceptual but it is something materially done. A concert hall, for example, uses walls, doors, and acoustics to physically separate the performance inside from the urban space outside, thereby creating a localized site for a specific kind of interaction.

Thirdly, AI is often granted an exceptional status in comparison to prior technologies. This framing posits GenAI as a uniquely ‘active’ agent, a quality not ascribed to other artifacts which are conversely relegated to a ‘passive’ role. This distinction is exemplified by Haase and Pokutta, who contrast GenAI systems with traditional Creativity Support Systems such as Photoshop; they argue that while they ‘facilitate creativity,’ they ‘remain tools rather than active contributors to the creative process’ (2024). Yet, attributing creative agency solely to AI systems is misleading (Bown, 2021) and “we should ask how agency and creativity are distributed in the network of actors who contribute to the process in question” (Moruzzi, 2022, p. 14). On the other hand, STS developed a broader understanding of technological agency stressing how the capacity to act and shape outcomes is not exclusive to humans or AI systems: agency should be seen in the various artifacts that participate in practices, which are not mere passive tools for human intentions but active contributors and mediators to what actors do (Latour, 2005; Hennion, 2020).

Finally, creativity is typically addressed with an ontological approach. As Celis Bueno et al. observe, these “approaches begin by asking ‘what is creativity’, and once defined, proceed to evaluate whether a specific individual or machine fits under this definition” (2024, p. 2). This is particularly evident in the field’s reliance on cognitive definitions of creativity and its attempts to create fixed taxonomies for the possible interactions between humans and AI. However, STS scholarship has demonstrated that agency and specifically creative agency is not an inherent property but is actively assigned to certain actors and actions through social, cultural, and technical processes (Micheal, 2017). The attribution of creativity is therefore a multiple and contested process. Ontologi-

cal approaches, which seek a single, fixed definition, are ill-equipped to capture this dynamic multiplicity. Specific valuation practices are performed by a diverse field of ‘information intermediaries’ who collectively shape who or what is deemed creative such as expert critics, awards, ratings and rankings (Sharkey et al., 2023). While certain accounts may become stabilized and transformed into standards, an STS perspective suggests that rather than accepting these valuation practices as given, we must investigate their sociomaterial construction showing how they shape who and what is included in or excluded from the category of ‘creative’.

Following Celis Bueno et al. (2024), we move beyond the ontological inquiry of ‘what co-creativity is’. Instead, we propose an empirical investigation into ‘where and how creativity is practically accomplished’ within situated creative practices. We hold that creativity is distributed and cannot be located neither in a specific actor (human or non-human), nor just in a process (cognitive or not), nor in a unique moment in space and time (Glaveanu, 2014). Recognizing creativity as distributed across space, time, and actors requires shifting focus toward an empirical study of the practices that involve specific human/non-human networks.

Application Domain and Historical Context

To investigate the agency and practices of Filmmaking we shall direct our analytical attention to the sites of production, showing how a film ‘comes into being’ through precise sociomaterial processes. Farías and Wilkie suggest that creativity investigations should look at the ‘humdrum and habitual workplace’ where creation happens (2016). The sites are not merely a physical space but a dense sociomaterial arrangement where practitioners ‘engage in conceiving, modelling, testing and finishing actual cultural artefacts. This approach reveals three dimensions of distribution: agency, space, and time.

First, film’s creative agency is distributed across a vast network of mediators both human—e.g. cinematographers, editors or crew members—and non-human—e.g. the specific camera and lenses used, the editing software. Film criticism often celebrates directors as the authors of the

film, an approach that draws on traditional film theories (Wollen, 2013). STS would rather consider this attribution a form of Boundary Work (Gieryn, 1983), a practice that traces boundaries between different domains of labor to assign specific responsibilities and legal rights. According to Caldwell (2013), this sustains a rigid and uneven division between above-the-line and below-the-line workers, where the latter are usually stripped of their creative agency in public discourse.

Secondly, in Filmmaking the site of production is typically displaced and distributed across multiple spaces, each dedicated to a specific activity: pre-production offices where scripts are written and storyboards are drawn, shooting locations, soundstages, and post-production houses where editing, sound design, and visual effects are completed. Therefore, the studio is not a single container for creativity but is itself a distributed network of sites, each contributing to the process.

Finally, Filmmaking is also characterized by a temporal distribution. The final cut of a film looks like a stable artifact but is the outcome of a long process involving many intermediary objects such as multiple script drafts, daily rushes, rough cuts, and discarded scenes. Each of these temporary objects enables interactions that eventually shape and influence the creative process in non linear ways (Pellegrinelli & Parolin, 2020). Relevant intermediary objects are what media production terms media assets, that is all intermediary materials that will finally be manipulated and assembled in the editing process—e.g., raw footage, sound recordings, and metadata. By tracing the effects of GenAI on media assets, our taxonomy will reveal the potential transformations of production processes.

To investigate and analyze the reconfigurations GenAI will bring to Filmmaking we should draw on the knowledge we can gain from the study of past technological innovations, which, from an STS approach, can be understood as processes that reconfigure pre-existing spatio-temporal interactions between human and non-human actors. We argue that each technology adopted in Filmmaking ultimately affects the final aesthetic outcome by altering the underlying production practices. This follows the work

of film scholars like Salt (2009), Beach (2015), and Bordwell (2018), which demonstrates that technical constraints and affordances have consistently served as primary drivers of artistic production and cinematic style. Also, film history often highlights the transition to sound, the introduction of color, and the digital turn as the medium's primary leaps. Peter Wollen (1980) critiqued this reductive view of technological development, noting that significant yet mundane innovations often remain overlooked. A prime example is the introduction of adhesive tape to the editing room, which fundamentally transformed the physical and creative labor of cutting film.

In the last decade, the advent of digital significantly reshaped Filmmaking practices leading toward what Caldwell calls 'blurred and collapsed workflows' (2013). By this, he refers to the way digital integration merges previously distinct stages of production—such as cinematography, visual effects, and editing—into a simultaneous and overlapping process. Swords and Willment (2024), for instance, have shown that the introduction of virtual production significantly challenges traditional production models, allowing post-production possibilities to dictate pre-production and production working practices.

Taxonomy construction

Drawing on the historical precedents discussed in §3.2, we contend that the integration of GenAI will likely trigger analogous sociomaterial reconfigurations in Filmmaking, even if current empirical research is not yet sufficient to fully map these consequences. It remains unclear whether AI represents a continuity of the trajectory of digitalization or a disruptive phenomenon that will radically restructure production leading to radical replacement or displacement of workers (Erickson, 2024).

Nonetheless, we propose an analytical lens to identify exactly where and how these processes can be studied. By focusing on the functional application of GenAI techniques, we can map their influence on established workflows without resorting to speculative scenarios. Instead, we explore the consequences of GenAI by focusing on the manipulation of media assets—images, sounds and vid-

eos—and ground the discussion in the concepts of distributed creativity, socio-material practices, and aesthetic effects. Within a distributed framework, media assets function as intermediary objects: they are the fundamental building blocks assembled to produce a film, and their creation is the result of coordinated interactions among diverse professional roles. Consequently, analyzing how specific AI techniques transform these assets allows us to trace the resulting shifts in professional practice.

The taxonomy is structured into three primary families of AI techniques based on their role in the asset production workflow: Asset Enhancement, Asset Editing, and Asset Generation. To evaluate the sociomaterial implications of these families, our analysis examines:

- the features of the different techniques and some uses in production;
- the ways in which their integration reconfigures established working roles and collaborative workflows;
- the potential aesthetic consequences these techniques have on the final film.

Results

In this section, we present the findings of our taxonomy and Table 1 provides a partial inventory of the families. Only three techniques are discussed here to demonstrate the taxonomy's structural logic, while a more comprehensive mapping is the subject of a recent paper (Masai et al., 2026).

Asset Enhancement

Asset enhancement refers to techniques designed to improve the quality, resolution, or specific attributes of existing media assets. As shown in Table 1, enhancement techniques like upscaling do not merely replace traditional tools but redistribute agency across the production timeline. By allowing close-ups to be digitally extracted from wider shots in post-production, these tools could alleviate the immediate pressure for precise framing during principal photography. This sociomaterial shift might grant actors greater freedom of movement and could optimize production time by reducing the need for frequent lens changes and lighting adjustments on set.

Tab. 1 - Asset enhancement techniques. Source: Masai, Carta, Lis (2026).

Family	Technique (examples)	Reconfiguration of practices	Aesthetic effects
Asset Enhancement	Upscaling: Increase of video footage resolution	• Less need to pay attention to composition in framing while shooting on set, saving time. • Decoupling camera resolution from final output via post-production enhancement. • Less need to change lenses on set to get a closer framing of the subjects: one could obtain a close-up from a long or medium shot.	• New styles of découpage • Different video quality standards
Asset Editing	Face transfer: Digital face replacement	• Stand-ins for stunts and synthetic reshoots	• New découpage choices when using stunt doubles
Asset Generation	Text-to-video: Synthetic production of assets	• Resolving on-set omissions via post-production synthesis of cut-away shots to avoid reshoots. • Synthesizing high-budget establishing shots and specialized environments (e.g., underwater sequences) without physical production costs.	• Unnatural images, movements and effects. • Unreal transitions • Non-physical photographic perspectives

Asset Editing

Asset editing comprises techniques that modify, rearrange, or replace specific elements within existing audiovisual materials. As illustrated in Table 1, these tools—such as digital face replacement—fundamentally reconfigure the relationship between physical performance and the final cinematic asset. In practice, this allows for the decoupling of a stunt double’s physical labor from the actor’s facial identity, eliminating the need for complex on-set makeup or staging constraints traditionally used to conceal stand-ins. Consequently, creative agency is mediated through a synthetic layer, enabling directors to make *découpage* choices that were previously impossible due to the physical limitations of the shoot.

Asset Generation

Asset generation involves the creation of new media content from scratch, typically through the use of text-based or visual prompts. This family of techniques represents a profound temporal redistribution of Filmmaking labor. By utilizing tools like text-to-video synthesis, editors can generate missing continuity assets—such as a specific cut-away or a high-cost establishing shot—directly during the post-production phase. This resolves on-set omissions and bypasses the necessity for physical reshoots. While this collapses traditional workflows, it intro-

duces new aesthetic possibilities where the visual product is no longer strictly bound to the spatial or temporal constraints of principal photography.

Conclusions

With this paper we challenged the ‘co-creativity’ paradigm by proposing a shift toward a distributed creativity approach and reframed GenAI not as an autonomous collaborator, but as an actor within a broader sociomaterial network. Hence, we moved beyond speculative binaries of human replacement versus augmentation, focusing instead on the practical reconfigurations of creative labor. Through the proposed taxonomy, we have demonstrated that GenAI techniques could redefine media asset production and consequently the spatio-temporal dynamics of Filmmaking. Our analysis shows that while AI introduces new affordances, it follows a historical trajectory where technical constraints have consistently influenced aesthetic and professional shifts. Future research should prioritize empirical studies of these evolving workflows to further map how the category of ‘creative work’ is being redefined. By investigating ‘creativity in the making’, we can better understand the political and aesthetic stakes of AI integration in the media industries.

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References

- Beach, C. (2015). *A Hidden History of Film Style: Cinematographers, Directors, and the Collaborative Process*. Univ of California Press.
- Bordwell, D. (2018). *On the History of Film Style*. Second Edition. Irvington Way Press. <https://www.davidbordwell.net/books/onthehistory.php>
- Bown, O. (2021). *Beyond the Creative Species: Making Machines That Make Art and Music*. MIT Press.
- Caldwell, J. T. (2013). Authorship Below-the-Line. In Gray, J., & Johnson, D. (Eds.). *A Companion to Media Authorship*. John Wiley & Sons Ltd, pp. 347–369. <https://doi.org/10.1002/9781118505526.ch18>
- Callon, M. (1998). An Essay on Framing and Overflowing: Economic Externalities Revisited by Sociology. *The Sociological Review*, 46(1_suppl), 244–269. <https://doi.org/10.1017/S0038013298000000>

- org/10.1111/j.1467-954X.1998.tb03477.x
- Celis Bueno, C., Chow, P.-S., & Popowicz, A. (2024). Not “what”, but “where is creativity?”: Towards a relational-materialist approach to generative AI. *AI & SOCIETY*, 40(2), 339–351. <https://doi.org/10.1007/s00146-024-01921-3>
- Davis, N., Sherson, J., & Rafner, J. (2025). The Co-Creative Design Framework for Hybrid Intelligence. In *Proceedings of the 2025 Conference on Creativity and Cognition*, pp. 560–572. <https://doi.org/10.1145/3698061.3726934>
- Erickson, K. (2024). AI and work in the creative industries: Digital continuity or discontinuity? *Creative Industries Journal*, 1–21. <https://doi.org/10.1080/17510694.2024.2421135>
- Fariás, I., & Wilkie, A. (Eds.). (2015). *Studio Studies: Operations, Topologies & Displacements*. Routledge. <https://doi.org/10.4324/9781315756523>
- Gieryn, T. F. (1983). Boundary-work and the demarcation of science from non-science: Strains and interests in professional ideologies of scientists. *American Sociological Review*, 781–795.
- Haase, J., & Pokutta, S. (2024). *Human-AI Co-Creativity: Exploring Synergies Across Levels of Creative Collaboration* (arXiv:2411.12527). arXiv. <https://doi.org/10.48550/arXiv.2411.12527>
- Hennion, A. (2020). *The passion for music: A sociology of mediation* (M. Rigaud & P. Collier, Trans.; First issued in paperback). Routledge, Taylor & Francis Group.
- Kantosalo, A., & Takala, T. (2020). Five C’s for Human–Computer Co-Creativity—An Update on Classical Creativity Perspectives. In *Iccc*, pp. 17–24.
- Latour, B. (1996). On Interobjectivity. *Mind, Culture, and Activity*, 3(4), 228–245. https://doi.org/10.1207/s15327884mca0304_2
- Latour, B. (2005). *Reassembling the social: An introduction to Actor-Network-Theory*. Oxford Univ. Press.
- Lin, Z., Ehsan, U., Agarwal, R., Dani, S., Vashishth, V., & Riedl, M. (2023). Beyond Prompts: Exploring the Design Space of Mixed-Initiative Co-Creativity Systems. *arXiv*. (arXiv:2305.07465). <https://doi.org/10.48550/arXiv.2305.07465>
- Masai, P., Carta, L., & Lis, M. M. (2026). Integrating GenAI in Filmmaking: From Co-Creativity to Distributed Creativity. *arXiv*. <https://arxiv.org/abs/2603.23415>
- Michael, M. (2017). *Actor-network theory: Trials, trails and translations*. Sage.
- Moruzzi, C. (2022). Creative Agents: Rethinking Agency and Creativity in Human and Artificial Systems. *Journal of Aesthetics and Phenomenology*, 9(2), 245–268. <https://doi.org/10.1080/20539320.2022.2150470>
- Parolin, L. L., & Pellegrinelli, C. (2020). Unpacking distributed creativity: Analysing sociomaterial practices in theatre artwork. *Culture & Psychology*, 26(3), 434–453. <https://doi.org/10.1177/1354067X19894936>
- Rafner, J. (2025). Computational Co-Creativity. In J. Katz-Buonincontro & T. Kettler (Eds.), *The Oxford Handbook of Creativity and Education*. Oxford University Press, pp. 817–845. <https://doi.org/10.1093/oxfordhb/9780197698181.013.0043>
- Rezwana, J., & Maher, M. L. (2022). Designing Creative AI Partners with COFI: A Framework for Modeling Interaction in Human-AI Co-Creative Systems. *ACM Transactions on Computer-Human Interaction*, 30(5), 1–28. <https://doi.org/10.1145/3519026>
- Salt, B. (2009). *Film style and technology: History and analysis* (Third edition). Starword.
- Sharkey, A., Kovács, B., & Hsu, G. (2023). Expert Critics, Rankings, and Review Aggregators: The Changing Nature of Intermediation and the Rise of Markets with Multiple Intermediaries. *Academy of Management Annals*, 17(1), 1–36. <https://doi.org/10.5465/annals.2021.0025>
- Swords, J., & Willment, N. (2024). ‘It used to be fix-it in post production! now it’s fix-it in pre-production’: How virtual production is changing production networks in film and television. *Creative Industries Journal*, 1–17. <https://doi.org/10.1080/17510694.2024.2430025>
- Wollen, P. (1980). Cinema and Technology: A Historical Overview. In T. de Lauretis & S. Heath (Eds.), *The Cinematic Apparatus*. Palgrave Macmillan UK, pp. 14–25. https://doi.org/10.1007/978-1-349-16401-1_2
- Wollen, P. (2013). *Signs and Meaning in the Cinema*. Bloomsbury Academic.